Junko Munakata Marr, Professor of Civil and Environmental Engineering

Alexander Pak, Assistant Professor, Chemical and Biological Engineering

Steve Pankavich, Department Head and Professor of Applied Mathematics & Statistics

Anthony Petrella, Department Head and Professor of Mechanical Engineering

Yamuna Phal, Assistant Professor of Electrical Engineering

Matthew Posewitz, Professor of Chemistry

James Ranville, Professor of Chemistry

Jonathan Sharp, Associate Department Head and Professor of Civil and Environmental Engineering

Anne Silverman, Professor of Mechanical Engineering

E. Dendy Sloan, Emeritus Professor of Chemical and Biological Engineering

John Spear, Professor, Civil and Environmental Engineering

Jeff Squier, Professor of Physics

Amadeu Sum, Professor of Chemical and Biological Engineering

Brian Trewyn, Department Head and Professor of Chemistry

Shubham Vyas, Professor of Chemistry

Xiaoli Zhang, Professor of Mechanical Engineering

Teaching Faculty

Linda Battalora, Teaching Professor of Petroleum Engineering

Suzannah Beeler, Associate Director of the QBE Undergraduate Program and Teaching Associate Professor of Chemical and Biological Engineering

Christian Beren, Director of the QBE Undergraduate Program and Teaching Associate Professor of Chemistry

Kristine Csavina, Teaching Professor of Mechanical Engineering

Duncan Davis-Hall, Teaching Assistant Professor of Engineering, Design, & Society

Alina Handorean, Teaching Professor of Engineering, Design & Society

Jenny Ketterling, Teaching Associate Professor of Chemical and Biological Engineering

Josh Ramey, Teaching Associate Professor of Chemical and Biological Engineering

Justin Shaffer, Associate Dean of Undergraduate Students and Teaching Professor of Chemical and Biological Engineering

Quantum Engineering

Degrees Offered

- Graduate certificate in Quantum Engineering
- Master of Science (non-thesis)
- Master of Science (thesis)

Program Description

Quantum Engineering is an interdisciplinary program that seeks to equip students for careers in emerging technologies based on quantum entanglement. It encompasses a wide range of disciplines, including physics, materials science, computer science, electrical engineering, chemistry, and mathematics, and is a collaborative effort among many Mines departments.

Two Master's degrees and one professional certificate are offered. Quantum Engineering has two tracks for both degrees and the professional certificate.

- Quantum Engineering Hardware (QEH) track focuses on experimental techniques relevant to quantum technology
- Quantum Engineering Software (QES) track focuses on theory, algorithms, and simulation

Primary Contact

Quantum Engineering Program
https://quantum.mines.edu/

Program Requirements

Quantum Engineering is an interdisciplinary program that seeks to equip students for careers in emerging technologies based on quantum entanglement. It encompasses a wide range of disciplines that include physics, materials science, computer science, and mathematics, and is necessarily a collaborative effort among many Mines departments. Two master's degrees and one graduate certificate are offered.

For both degrees and the graduate certificate, Quantum Engineering has two "tracks" as summarized below. The Quantum Engineering Hardware (QEH) track will focus on experimental techniques relevant to quantum technology, while the Quantum Engineering Software (QES) track will focus on theory, algorithms and simulation. Students must choose a track to complete the program, but they may take courses from both tracks provided they meet the prerequisite requirements.

MS Degree Curriculum Requirements:

A Master of Science in Quantum Engineering will consist of 30 total credits. Students may choose a thesis or non-thesis option for this degree. For the thesis option, 9 credits out of the 30 are devoted to thesis research leading to an acceptable Master's thesis. Students choosing the non-thesis option will devote all 30 credits to coursework. Regardless of the option chosen, 9 of the coursework credits will be devoted to the required core classes for the chosen track.

Reflecting the interdisciplinary nature of the program, we strongly recommend to our students that at least 9 total credits of the MS degree coursework should come from courses in a department outside of the student's undergraduate major. The required core courses, if outside of the student's major, would count toward this recommendation. Our guiding philosophy is that the problem of building a quantum computer is a complex, interdisciplinary one which requires contributions from a

vast array of subfields, and young scientists who appreciate this will likely have a far better perspective on the field than those who do not.

MS Non-Thesis Software Track

PHGN519	FUNDAMENTALS OF QUANTUM INFORMATION	3.0
CSCI581	QUANTUM PROGRAMMING	3.0
PHGN545	QUANTUM MANY-BODY PHYSICS	3.0
PHGN526	QUANTUM ENGINEERING SEMINAR (FALL)	1.0
PHGN527	QUANTUM ENGINEERING SEMINAR (SPRING)	1.0
Electives		19.0
Total Semester Hrs		**30.0**

MS Non-Thesis Hardware Track

PHGN519	FUNDAMENTALS OF QUANTUM INFORMATION	3.0
EENG532	LOW TEMPERATURE MICROWAVE MEASUREMENTS FOR QUANTUM ENGINEERING	3.0
PHGN535	INTERDISCIPLINARY SILICON PROCESSING LABORATORY	3.0
PHGN526	QUANTUM ENGINEERING SEMINAR (FALL)	1.0
PHGN527	QUANTUM ENGINEERING SEMINAR (SPRING)	1.0
Electives		19.0
Total Semester Hrs		**30.0**

MS Thesis Software Track

PHGN519	FUNDAMENTALS OF QUANTUM INFORMATION	3.0
CSCI581	QUANTUM PROGRAMMING	3.0
PHGN545	QUANTUM MANY-BODY PHYSICS	3.0
PHGN526	QUANTUM ENGINEERING SEMINAR (FALL)	1.0
PHGN527	QUANTUM ENGINEERING SEMINAR (SPRING)	1.0
Electives		9.0
PHGN707	GRADUATE THESIS / DISSERTATION RESEARCH CREDIT	10.0
Total Semester Hrs		**30.0**

MS Thesis Hardware Track

PHGN519	FUNDAMENTALS OF QUANTUM INFORMATION	3.0
EENG/PHGN532	LOW TEMPERATURE MICROWAVE MEASUREMENTS FOR QUANTUM ENGINEERING	3.0
PHGN535	INTERDISCIPLINARY SILICON PROCESSING LABORATORY	3.0
PHGN526	QUANTUM ENGINEERING SEMINAR (FALL)	1.0
PHGN527	QUANTUM ENGINEERING SEMINAR (SPRING)	1.0
Electives		9.0
PHGN707	GRADUATE THESIS / DISSERTATION RESEARCH CREDIT	10.0
Total Semester Hrs		**30.0**

Coursework Details:

QES students will be required to take these courses in the following sequence:

In the fall:

- PHGN519, Fundamentals of Quantum Information

In the spring:

- CSCI581, Quantum Programming
- PHGN545, Quantum Many-Body Physics

QEH students will be required to take these courses in the following sequence:

In the fall:

- PHGN519, Fundamentals of Quantum Information

In the spring:

- PHGN535, Interdisciplinary Silicon Processing Laboratory
- PHGN532, Low Temperature Microwave Measurements for Quantum Applications

Approved Electives:

Physics Electives		
PHGN511	MATHEMATICAL PHYSICS	3.0
PHGN520	QUANTUM MECHANICS I	3.0
PHGN521	QUANTUM MECHANICS II	3.0
PHGN530	STATISTICAL MECHANICS	3.0
PHGN550	NANOSCALE PHYSICS AND TECHNOLOGY	3.0
PHGN566	MODERN OPTICAL ENGINEERING	3.0
PHGN581	LASER PHYSICS	3.0
PHGN585	NONLINEAR OPTICS	3.0
Computer Science Electives		
CSCI542	SIMULATION	3.0
CSCI561	THEORY OF COMPUTATION	3.0
CSCI563	PARALLEL COMPUTING FOR SCIENTISTS AND ENGINEERS	3.0
CSCI564	ADVANCED COMPUTER ARCHITECTURE	3.0
CSCI571	ARTIFICIAL INTELLIGENCE	3.0
CSCI575	ADVANCED MACHINE LEARNING	3.0
CSCI574	THEORY OF CRYPTOGRAPHY	3.0
CSCI580	ADVANCED HIGH PERFORMANCE COMPUTING	3.0
Electrical Engineering Electives		
EENG509	SPARSE SIGNAL PROCESSING	3.0
EENG517	THEORY AND DESIGN OF ADVANCED CONTROL SYSTEMS	3.0
EENG526	ADVANCED ELECTROMAGNETICS	
EENG528	COMPUTATIONAL ELECTROMAGNETICS	
EENG530	PASSIVE RF & MICROWAVE DEVICES	3.0
EENG533	ACTIVE RF & MICROWAVE DEVICES	
EENG617	INTELLIGENT CONTROL SYSTEMS	3.0
EENG618	NONLINEAR AND ADAPTIVE CONTROL	3.0
Metallurgy and Material Engineering Electives		
Materials Science Electives		
MTGN605	ADVANCED TRANSMISSION ELECTRON MICROSCOPY	2.0

MTGN605L	ADVANCED TRANSMISSION ELECTRON MICROSCOPY LABORATORY	1.0
MTGN656	ADVANCED ELECTRON MICROSCOPY	2.0
MTGN656L	ADVANCED ELECTRON MICROSCOPY LABORATORY	1.0
MLGN502	SOLID STATE PHYSICS	3.0
MTGN573	COMPUTATIONAL MATERIALS	3.0
MLGN515	ELECTRICAL PROPERTIES AND APPLICATIONS OF MATERIALS	3.0
MLGN583	PRINCIPLES AND APPLICATIONS OF SURFACE ANALYSIS TECHNIQUES	3.0
MLGN593	BONDING, STRUCTURE, AND CRYSTALLOGRAPHY	3.0
Applied Mathematics and Statistics Electives		
MATH506	COMPLEX ANALYSIS II	3.0
MATH510	ORDINARY DIFFERENTIAL EQUATIONS AND DYNAMICAL SYSTEMS	3.0
MATH536	ADVANCED STATISTICAL MODELING	3.0
MATH538	STOCHASTIC MODELS	3.0
MATH550	NUMERICAL SOLUTION OF PARTIAL DIFFERENTIAL EQUATIONS	3.0
MATH551	COMPUTATIONAL LINEAR ALGEBRA	3.0
Humanities, Arts, and Social Sciences Electives		
HASS523	ADVANCED SCIENCE COMMUNICATION	3.0

Mines' Combined Undergraduate/Graduate Degree Program:

Students enrolled in Mines' combined undergraduate/graduate program may double count up to 6 credits of graduate coursework to fulfill requirements of both their undergraduate and graduate degree programs. These courses must have been passed with B- or better, not be substitutes for required coursework, and meet all other university, department, and program requirements for graduate credit.

Students are advised to consult with their undergraduate and graduate advisors for appropriate courses to double count upon admission to the combined program.

Graduate Certificate Curriculum Requirements:

The certificate option consists of three of the four new courses, plus one additional elective chosen from the above list, for a total of 12 credits.

Graduate Certificate, Software Track

PHGN519	FUNDAMENTALS OF QUANTUM INFORMATION	3.0
CSCI581	QUANTUM PROGRAMMING	3.0
PHGN545	QUANTUM MANY-BODY PHYSICS	3.0
Elective		3.0
Total Semester Hrs		**12.0**

Graduate Certificate, Hardware Track

PHGN519	FUNDAMENTALS OF QUANTUM INFORMATION	3.0
PHGN535	INTERDISCIPLINARY SILICON PROCESSING LABORATORY	3.0
EENG/PHGN532	LOW TEMPERATURE MICROWAVE MEASUREMENTS FOR QUANTUM ENGINEERING	3.0
Elective		3.0
Total Semester Hrs		**12.0**

Program Director

Meenakshi Singh, Associate Professor, Physics

Department of Chemical and Biological Engineering

Matthew Crane, Assistant Professor

Department of Computer Science

Neil Dantam, Assistant Professor

Dinesh Mehta, Professor

Bo Wu, Associate Professor

Dejun Yang, Associate Professor

Department of Electrical Engineering

Peter Aaen, Professor

Department of Metallurgical and Materials Engineering

Geoff Brennecka, Professor

Brian Gorman, Professor

Department of Physics

Victoria Borish, Teaching Assistant Professor

Lincoln Carr, Professor

Patrice Genevet, Professor

Zhexuan Gong, Assistant Professor

Eliot Kapit, Associate Professor

Wouter Van De Pontseele, Assistant Professor

Frederic Sarazin, Professor

Jeff Squier, Professor

Affiliated Faculty

Matt Beard, Joint Appointment, NREL and Chemistry

Justin Johnson, NREL

Andriy Zakutayev, Joint Appointment, NREL and Metallurgical and Materials Engineering

Robotics

Degrees Offered

- Graduate certificate in Robotics
- Master of Science in Robotics (non-thesis)
- Master of Science in Robotics (thesis)
- Doctor of Philosophy in Robotics

Program Description

The Robotics program offers a graduate certificate and the degrees of Master of Science and Doctor of Philosophy in Robotics. The graduate certificate is intended for working professionals. The non-thesis MS is designed to prepare candidates for industry careers in robotics. The thesis MS and PhD degrees are designed to prepare students for research careers.

Primary Contact

Robotics Program
https://robotics.mines.edu/

Program Director

Xiaoli Zhang

Professors

Qi Han

Kathryn Johnson

Tyrone Vincent

Xiaoli Zhang

Associate Professors

Hao Zhang

Assistant Professors

Micah Corah

Neil Dantam

Kaveh Fathian

George Kontoudis

Yangming Shi

Thomas Williams

Qin Zhu

Frankie Zhu

Professor Emeritus

Kevin Moore

The Robotics program offers graduate certificate, Master of Science and Doctor of Philosophy degrees in Robotics. The graduate certificate is intended for working professionals. The non-thesis MS is designed to prepare candidates for industry careers in robotics. The thesis MS and PhD degrees are designed to prepare students for research careers.

Combined Program: The Robotics program also offers combined BS+MS degrees. These degrees offer an expedited graduate school application process and allow students to begin graduate coursework while still finishing their undergraduate degree requirements.

Admissions and Program Policies

Mines' Combined Undergraduate/Graduate Degree Program

Students enrolled in Mines' combined undergraduate/graduate program may double count up to 6 credits of graduate coursework to fulfill requirements of both their undergraduate and graduate degree programs. These courses must have been passed with B- or better, not be substitutes for required coursework, and meet all other university, department, and program requirements for graduate credit.

Students are advised to consult with their undergraduate and graduate advisors for appropriate courses to double count upon admission to the combined program.

MS and PhD

The minimum requirements for admission to the MS and PhD degrees in Robotics are:

• Applicants must have a Bachelor's degree, or equivalent, from an accredited institution with a grade-point average of 3.0 or better on a 4.0 scale prior to matriculating into the Robotics degree program.

• Students are expected to have completed the following coursework: 1) two semesters of calculus, 2) differential equations, and 3) data structures. The Robotics graduate admissions committee may require that students who do not meet this expectation demonstrate competency or take remedial coursework. Such coursework may not count toward the graduate degree. The committee will decide whether to recommend regular or provisional admission.

• Graduate Record Examination (Quantitative section) score of 151 or higher (or 650 on the old scale). Applicants who have graduated with a computer science, engineering, or math degree from Mines within the past five years are not required to submit GRE scores.

• Applicants who are not U.S. citizens or permanent residents and whose native language is not English, must prove English proficiency as part of the application process. Refer to the Graduate Admissions website for more details.

• For the PhD program, prior research experience is desired but not required.

Transfer Credit

Graduate-level courses taken at other universities for which a grade equivalent to a B or better was received will be considered for transfer credit with approval of the advisor and/or thesis committee, and home department head, as appropriate. Transfer credits must not have been used as credit toward a bachelor's degree. For the MS degree, no more than nine credits may transfer. For the PhD degree, up to 24 credits of courses may be transferred. In lieu of transfer credit for individual courses, students who enter the PhD program with a thesis-based master's degree from another institution may transfer up to 36 credits in recognition of the course work and research completed for that degree.